NAS-Guided Lightweight Multiscale Attention Fusion Network for Hyperspectral Image Classification

Jianing Wang, *Member, IEEE*, Runhu Huang, Siying Guo, Linhao Li, Minghao Zhu, Shuyuan Yang, *Senior Member, IEEE*, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— Deep learning (DL) has become a hot topic in the research field of hyperspectral image (HSI) classification. However, with increasing depth and size of deep learning methods, its application in mobile and embedded vision applications has brought great challenges. In this article, we address a network architecture search (NAS)-guided lightweight spectral–spatial attention feature fusion network (LMAFN) for HSI classification. The overall architecture of the proposed network is guided by several conclusions of NAS, which achieves fewer parameters and lower computation cost with deeper network structure by exploiting multiscale Ghost grouped with efficient channel attention (ECA) module for adaptively adjusting the weights of different channels. It helps fully extract spectral–spatial discriminant features to avoid information loss of the dimension reduction operation. Specifically, a multilayer feature fusion method is proposed to extract the fusion information of the spectral–spatial features of each layer by considering complementary information of different hierarchical structures. Therefore, high-lever spectral–spatial attributes are gradually exploited along with the increase in layers and the fusion of layers. The experimental verification on three real HSI data sets demonstrates that the proposed framework presents more satisfying classification performance and efficiency with deeper network structure and lower parameter size.**

***Index Terms*— Attention mechanism, hyperspectral image (HSI) classification, lightweight structure, multilayer feature fusion, multiscale convolution.**

I. Introduction

HYPERSPECTRAL image (HSI) system can obtain rich spectral and spatial information about the landcover surface. Therefore, the HSI can be viewed as a three-dimensional (3-D) digital cube that consists of hundreds of continuous narrow spectral bands spanning the visible to infrared wavelengths by hyperspectral remote sensors, which makes it possible to identify and distinguish materials accurately. In the past decade, analyzation and processing of HSI have been predominantly used in agriculture [1], environmental science [2], urbanization analysis [2], military defense [3], and other fields [4].

Numerous effective classification methods have been developed for HSI classification, most of the conventional methods utilized hand-crafted features are modeled and combined with a classifier for HSI classification. In [5], support vector machines (SVM), with nonlinear kernel projection method is proposed. However, it presents unsmooth classification result due to only utilizing spectral feature extraction for HSI classification. In order to effectively explore the spatial information with spectral features, SVM with composite kernel (SVMCK) [6], sparse representation-based classifie (SRC) [7], spatial-spectral derivative-aided kernel joint sparse representation (KJSR-SSDK) [8], and adaptive nonlocal spatial–spectral kernel (ANSSK) [9] introduced spatial features into HSI classification for extracting spectral–spatial combination features. Most of the aforementioned conventional algorithms are based on hand-crafted features and conventional learning models, while hand-crafted features usually require professional domain knowledge and they are difficult to address all the detailed requirements in data feature extraction.

Recently, deep learning-based algorithms have aroused great attention for remote sensing processing [10]–[13]. Among the HSI classification methods studied in the literature in recent years, the HSI classification methods based on deep learning have received great attention and exhibited good classification performance. Advances in joint spectral–spatial feature encoding approach demonstrates that spatial structure can provide extra contexture information for improving classification accuracies as well as smoothing classification results. Zhang *et al.* [14] and Chen *et al.* [15] first applied unsupervised deep feature learning, including stacked autoencoders (SAEs) and deep belief networks (DBNs) for spectral–spatial feature extraction and classification. Chen *et al.* [16], introduced supervised 2 dimensional convolutional Neural Network (2D-CNN), and 3 dimensional convolutional Neural Network (3D-CNN), where 2D-CNN constructed by three 2-D convolutional layers, and 3D-CNN consists of three layers of 3-D convolution for HSI classification. In order to augment the available training samples for the training procedure of CNN, Li *et al.* [17] proposed CNN pixel-pair feature (CNN-PPF)

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The authors are with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education of China, School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: circuitwang@163.com).

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method by utilizing pixel-pair combination to increase the number of training samples for improving the classification performance. He *et al.* [18] introduced the multiscale 3D-CNN by considering filters with varying size.

Recent application performance evidences that deeper neural networks present more satisfying result for many visual recognition tasks, whereas the deeper neural network also increased the instability of the network and the difficulty of optimization. Instead of learning unreferenced functions, by reformulating the layers as learning residual functions concerning the layer inputs, the residual network is introduced in [19] to gain accuracy and stability of optimization. Zhong *et al.* [20] migrated the ResNet model to spectral–spatial CNN-based algorithm (SSRN), which presents a satisfying result for improving HSI classification performance. Meanwhile, to reduce the problems of overfitting and gradient disappearance in deep CNNs by considering the strong complementary correlated information among different hierarchical layers of the neural network, deep feature fusion network (DFFN) is proposed in [21]. However, aforementioned models usually cannot get promising result with deeper layers as increasing the numbers of layer-wise trainable parameters.

In recent years, the lightweight network structure is proposed mainly to reduce network parameters and speed up network operation, which has attracted great attention from researchers [22]–[24]. Mainly two existing mainstreams for the lightweight algorithm: one is the model compression that compressing the existing large deep learning (DL) network with excellent performance; the other is to design a new efficient convolution operation for reducing training parameters and computational complexity with increased depth of the network. For example, the depthwise (DW) separable convolutions structure first appeared in [25] Mobilenet V1 [26] and Inception model [27], which decomposes standard convolution integral into DW convolution and 1×1 pointwise (PW) convolution. The method greatly reduces the amount of parameter usage and calculation. Whereafter, linear bottlenecks and inverted residual (Ir) blocks are introduced in the Mobilenet V2 model [28]. The former solves the problem that the weight of the output convolution kernel is empty, while the latter increases the dimension of feature extraction by 1×1 convolution kernel and the amount of information extraction. The latest Mobilenet V3 [29] utilizes architecture search technology, h-swish nonlinear activation function, and Squeeze-and-Excitation (SE) attention mechanism [30] for further reducing the number of network parameters as well as improving network performance. Shufflenet V1 [31] first reduces the computation of PW convolution by using group convolution with channel shuffle to circulate the spectral dimension feature information, which greatly reduces the computation of the model while maintaining the accuracy. Subsequently, Shufflenet V2 [32] further reduces the calculation cost by evaluating the direct metric on the target platform, beyond only considering floating point of operations (FLOPs). Additionally, factorized networks [33] are introduced by utilizing factorized convolution and topological connections. The DW separable filter is scaled up in Xception [34], and Squeezenet [35] designs a very small network using a bottleneck approach. Furthermore, Ghost [36], a plug-and-play lightweight structure, makes full use of redundant information in images designed for image classification. Chen *et al.* [37] proposed a novel extremely separated convolutional block (XSepConv), which fuses spatially separable convolutions into DW convolution to further reduce the computational cost and parameter size of large kernels. In addition, an extra 2×2 DW convolution coupled with improved symmetric padding strategy is used to compensate the negative effect brought by separable spatial convolution. XSepConv is designed as an effective alternative to ordinary deep convolution with large kernel sizes. Recently, lightweight networks have gained attention for HSI classification with limited sample sets. 3D-LWNET [38] is introduced cross-sensor and cross-mode transfer learning and lightweight structures into 3-D CNNs with expanding the previous 3-D CNN to dozens of convolutional layers. Roy *et al.* [39] proposed an S3EresBoF network framework, instead of fully connected layer, scale-invariant Bag-of-Features (BoF) is introduced to integrate with Resnet for subduing overfitting problem of the proposed structure.

Recently, attention mechanism has been widely used in deep CNNs and shows great potential in improving the performance of CNNs in terms of suppressing redundant information in the feature map and extracting effective features. SENet is an effective attention mechanism and is first proposed to suppress redundant feature maps by weighting channel direction feature maps. Convolutional block attention module (CBAM) [40] combines channel attention with spatial attention through pooling operation. NLNet [41] combines matrix multiplication operations with convolution operations to capture the long-range relationship in the global space. Combining NLNet and SENet, Cao *et al.* [42] proposed a simplified lightweight module GCNet for effectively extracting global context. Ma *et al.* [43] presented an deep connected attention network (DCANet) that interconnects adjacent attention blocks and trained jointly for making information interaction among attention blocks. Misra *et al.* [44] investigated lightweight but effective attention mechanisms and presented triplet attention, a novel method for computing attention weights by capturing cross-dimension interaction using a three-branch structure. ECANet [45] improves the SE structure, which significantly reduces the complexity of the model and proposes a local cross-channel interaction strategy without dimension reduction. In order to overcome the problems of over-parameterization and inconsistent class ratio per mini-batch, Zheng *et al.* [46] created the S3NET framework based on a new lightweight network and a stratified sample-based training strategy, while the proposed spectral attention module as a soft band selection to refine redundant spectrum information.

With the application of HSI classification based on deep learning in edge computing and embedded devices presents a prospect in recent years, it is proved that the increase of model depth in CNNs is beneficial to the efficiency of the network. However, the deeper network usually gives rise to more parameters along with higher computation and difficulty for optimization of the whole network. Therefore, inspired by the above-mentioned problem and literature, in this arti-

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cle, we propose a network architecture search (NAS)-guided lightweight spectral–spatial fusion neural network (LMAFN) for HSI classification. According to the instructive experience of NAS proposed in [47] that networks with four stages usually perform better than those with three stages, and a large number of units in the third stage and a small number of units in the last stage usually indicates better performance, in the proposed LMAFN, PW convolution and the Ir unit are, respectively, used to extract spectral and spectral–spatial information. Meanwhile, skip residual connection is utilized for fully exchanging and extracting the feature in the model. In the Ir unit, PW convolution and Ghost modules are applied to replace the standard convolution for greatly reducing the parameters and computational complexity of the network, where multiscale Ghost module is used to generate more distinguishable feature maps through cheap operations. Meanwhile, efficient channel attention (ECA) module is used to adaptively adjust the channel weights for improving the ability of network feature extraction. Furthermore, the complementary and correlation information from different layers are considered by multilayer hierarchical feature fusion structure for improving the classification performance. Therefore, the proposed LMAFN framework has fewer parameters and lower computation cost with a deeper network structure by utilizing lightweight multiscale attention structures, which helps extract discriminant features by fully considering the spectral–spatial feature for eliminating information loss in dimension reduction operation. The experimental result further demonstrates that LMAFN achieves more satisfying classification result with fewer parameters and execution time cost.

The rest of this article is organized as follows. The second part introduces relevant methods used in this model. The third part introduces the network proposed. The fourth part shows the specific experiments and analysis. The fifth part summarizes this article and looks forward to the future work.

II. Background and Preliminaries

A. Convolution Operation

Denote $U^l \in \mathbb{R}^{S\times S\times C}$ as the input of convolution layer and generate the output feature map $U^{l+1} \in \mathbb{R}^{S\times S\times C'}$, where S is the spatial width and height of a square input feature map, C and C' are the number of channels in the input and output feature maps, respectively. $W^{l+1} \in \mathbb{R}^{K\times K\times C\times C'}$ indicates the convolution kernel of the $l+1$th convolution layer. K is the size of the convolution kernel, and C' is the number of convolution kernels.

1) Depthwise (DW) Separable Convolution: In the manually designed lightweight structure, the DW separable convolution is a classic lightweight structure. DW separable convolution is the solution of standard convolution integral into DW convolution and PW convolution, which can be regarded as the decomposition along with the feature channel domain.

The main aim of DW convolution is to extract the feature information on a single feature map that does not has interaction with each other. The function of PW convolution is to fuse the feature information obtained by the DW convolution operation that is not circulated with each other.

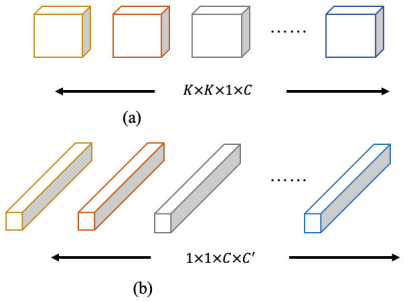


Fig. 1. DW separable convolution.

Fig. 1 indicates how to decompose the standard convolution into DW convolution and 1×1 PW convolution. Through the decomposition of convolution, the computational complexity and the number of parameters of the network are significantly reduced.

DW convolution with one filter per input channel (input depth) can be written as

$$\widehat{U}^{l+1}_{i,j,c} = \sum_{m=1}^{K}\sum_{n=1}^{K} \widehat{W}_{m,n,c} \cdot U^{l}_{i+m-1,j+n-1,c}. \tag{1}$$

The number of parameters required for standard convolution is calculated as follows:

$$P_{\mathrm{STD}} = K \times K \times C' \times C. \tag{2}$$

Standard convolutions have the computational cost of

$$C_{\mathrm{STD}} = K \times K \times C' \times C \times S \times S. \tag{3}$$

The number of parameters required for DW separable convolution is calculated as follows:

$$P_{\mathrm{DWS}} = C \times C' + K \times K \times C. \tag{4}$$

The calculation of this process is as follows:

$$C_{\mathrm{DWS}} = K \times K \times C \times S \times S + C \times C' \times S \times S. \tag{5}$$

By decomposing the standard convolution into a two-step process of filtering and combining, the reduced computation and parameters effect can be written as follows:

$$\frac{K \times K \times C \times S \times S + C \times C' \times S \times S}{K \times K \times C' \times C \times S \times S} = \frac{1}{C'} + \frac{1}{K^2}. \tag{6}$$

2) Ghost: In traditional deep convolution neural networks, standard convolution is the mainstream. It has the following two characteristics: First, take up a lot of computational costs; Second, generate redundant intermediate feature maps. To some extent, the introduction of DW separable convolution in MobileNet can reduce the computational cost. However, DW separable CNNs still have shortcomings that PW convolution occupies most of the computation in the DW separable CNNs.

Therefore, a novel Ghost module is proposed in [36] to generate more feature maps from cheap operations. Based on a set of intrinsic feature maps, Ghost uses a series of linear transformations at a low cost to generate many ghost feature maps that can completely reveal the information of inherent characteristics. Ghost module structure is shown in Fig. 2.

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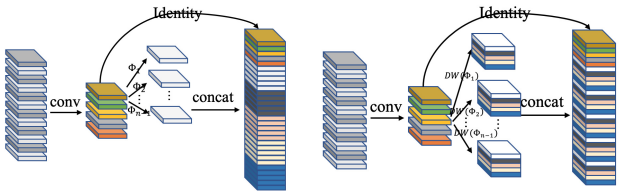


Fig. 2. Illustration of the proposed Ghost and implementation method. (a) Ghost. (b) Ghost implementation through DW.

As can be seen from Fig. 2(a), Ghost module is composed of standard convolution and Ghost. The output feature maps of convolutional layers often contain much redundancy as some of them could be similar to each other. It is unnecessary to spend a lot of parameters and FLOPs to generate these redundant feature maps. In Ghost module, standard convolution is utilized to reduce the feature dimension for removing the redundancy and then perform cheap multiscale linear mapping of a single feature map to generate more distinguishable features by using Ghost. Given the input feature map $U^l \in \mathbb{R}^{S\times S\times C}$ and the output feature map $U^{l+1} \in \mathbb{R}^{S\times S\times C'}$, where C and C' are the number of channels in the input and output feature maps. The calculation process of standard convolution is as follows:

$$\hat{U}^{l+1} = (W^l \otimes U^l) + b^l, \tag{7}$$

where $\otimes$ is the standard convolution operation. b^l indicates the value of bias of lth layer. $\hat{U}^{l+1}$ is the intrinsic feature map. $W^l \in \mathbb{R}^{K\times K\times C\times m}$ is the utilized filters, where $m \leq n$. The standard convolution here generally uses PW convolution. It has two main functions: First, it can remove the dimensionality reduction of feature maps; Second, compared with the DW separable convolution, the proportion of the PW convolution operation is reduced.

In order to further generate the required feature maps, apply a series of cheap linear operations on each intrinsic feature map in $\hat{U}^{l+1}$ to generate n ghost features:

$$U_{ij}^{l+1} = \Phi_{i,j}(\hat{U}_i^{l+1}) \quad \forall i = 1, 2, \ldots, m, \quad j = 1, 2, \ldots, n \tag{8}$$

where $\hat{U}_i^{l+1}$ is the ith intrinsic feature map in $\hat{U}^{l+1}$. $\Phi_{i,j}$ in the above function is the jth (except the last one) linear operation for generating the jth ghost feature map U_{ij}^{l+1}, that is, $\hat{U}_i^{l+1}$ has one or more Ghost feature maps $[U_{ij}^{l+1}]_{j=1}^n$. $\Phi_{i,j}$ can use multiscale convolution kernels to extract local information in different ranges, so more distinguishing features can be obtained. The last $\hat{U}_i^{l+1}$ has one or more ghost feature maps $[U_{ij}^{l+1}]_{j=1}^n$. $\Phi_{i,n}$ is the identity mapping for preserving the intrinsic feature maps as shown in Fig. 2(a). In the end, we can get $C' = m \times n$ feature maps.

Fig. 2(b) is an equivalent implementation form of Fig. 2(a). By utilizing multiple DW convolutions with multiscale convolution kernels, it realizes cheap linear operations for each intrinsic feature map and connects the identity mapping with the linear mapping to get the final output.

B. Attention Mechanism

DW separable convolution separates spatial feature fusion and the channel feature fusion that hampers the feature interaction between the channels. In order to facilitate the ability of DW separable convolution on channel dimension feature fusion, SE structure is introduced to involve and exchange the information between the channels so that the model can automatically learn the importance of different channel features [30]. Therefore, ECA, a lightweight module, empirically proves that the avoidance of dimension reduction with learnable channel attention can significantly reduce the complexity of the model while maintaining performance by appropriate cross-channel interaction.

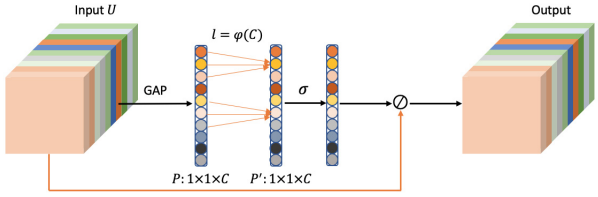


Fig. 3. U is the input feature map. P is the aggregated feature after the global average pooling (GAP). $\oslash$ represents the element-wise product.

ECA module is shown in Fig. 3, where ECA generates channel weights by performing a fast 1-D convolution of size l, where l is adaptively determined via a mapping of channel dimension C, and its calculation process is as follows [45]:

$$P_c = \mathrm{GAP}(U_c) = \frac{1}{S\times S}\sum_i^S\sum_j^S U_c(i, j) \tag{9}$$

where $U_c(i, j)$ is the value at the position (i, j) of the cth channel in the input feature map, P_c is the pooling value of the cth channel of the input feature map after the GAP, and l is the adaptive local cross-channel scope, the size of 1-D convolution kernel. The value of l is determined by the following equation:

$$l = \varphi(C) = \left|\frac{\log_2(C)}{\gamma} + \frac{b}{\gamma}\right|_{\mathrm{odd}} \tag{10}$$

where $|L|_{\mathrm{odd}}$ represents the nearest odd number of L, we set γ and b as 2 and 1, respectively. C is the number of channels. Therefore, when γ and b are determined, the value of l is only adaptively determined by channel C. The channel weight vector is finally calculated by the following formula:

$$\omega_c = \sigma\left(\sum_{j=1}^{l} W^j P_c^j\right), \quad P_c^j \in \phi_c^l \tag{11}$$

where σ represents the sigmoid function and ϕ_c^s indicates the set of l adjacent channels of P_c.

III. Proposed Framework

In this section, we first introduce the proposed HSI classification framework and two information extraction modules: SpeM and LW-SpaM. Meanwhile, ECA mechanism and MFFM utilized in the proposed framework are illustrated in detail. The loss function and optimization method are given in the end of this section.

A. Overview of the Proposed Framework

The overview of the proposed NAS-guided lightweight efficient multiscale attention fusion network (LMAFN) is illustrated in Fig. 4. The whole framework can be roughly divided

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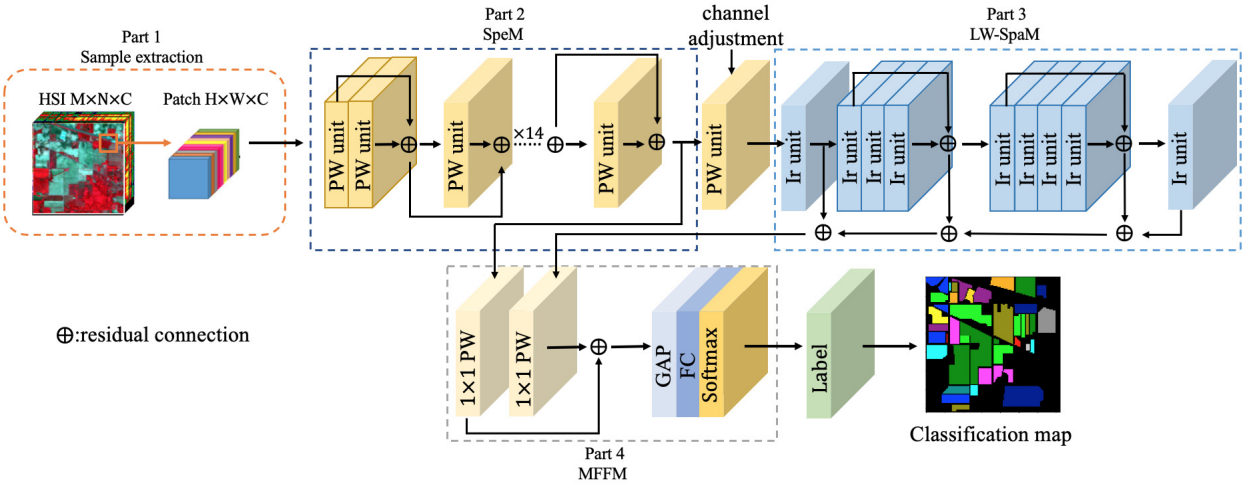


Fig. 4. Framework of the LMAFN for HSI classification. (1) Part 1 performs sample extraction. Split the original data set into patch input network for training. (2) Spectral feature extraction module (SpeM) extracts the spectral information of the input feature map through multilayer point convolution units. (3) Lightweight spatial feature extraction module (LW-SpaM) uses the lightweight Ghost with the ECA module to adjust channel attention to extract spatial information. (4) Multilayer feature fusion module (MFFM) helps to make full use of the feature information between different layers and further improves the classification effect.

into four parts, including sample extraction module, SpeM, lightweight spatial information extraction module (LW-SpaM), and MFFM. For sample extraction, suppose that the input HSI data set is $\mathcal{H} \in \mathbb{R}^{H\times W\times D}$, where H, W, and D are the height and the width of the spatial dimension and the number of the spectral dimension of HSI, respectively. $\mathcal{H}$ contains totally $H \times W$ pixels, including M labeled pixels $X = \{x_1, x_2, \ldots, x_M\} \in \mathbb{R}^{1\times 1\times D}$ and the number of pixel classes is C. The one-hot label vectors corresponding to the labeled pixel is $Y = \{y_1, y_2, \ldots, y_M\} \in \mathbb{R}^{1\times 1\times C}$. In order to make full use of the spectral and spatial information of HSIs, the pixel-centric 3-D patch cubes $Z = \{z_1, z_2, \ldots, z_M\} \in \mathbb{R}^{S\times S\times D}$ is extracted as the input, where S and D are the width and the number of channels of patch cubes, respectively. To make full use of spectral features in HSIs, we take all the spectral dimension information as input. For the 3-D patch cube z_i, the center pixel label of z_i is used as the label of the cube.

Through the first module of the proposed model, preprocessing $\mathcal{H}$ and the labeled 3-D patch cube data set is divided into Z_{train}, Z_{val}, and Z_{test} and corresponding label sets Y_{train}, Y_{val}, Y_{test}, respectively. For the SpeM, the module uses multiple 1×1 PW convolution units (PW unit) to reduce the dimension of the input 3-D patch cube and combine the feature information between channels through the PW convolution with low parameters and computation cost. In order to consider original spectral–spatial information, we utilized residual learning on SpeM, meanwhile the output of SpeM is added with spatial feature extraction module LW-SpaM under the instructive experience of NAS proposed in [47]. Therefore, we divide LW-SpaM into four stages with different numbers of lightweight Ir units, where each Ir unit is composed of PW separation convolution and Ghost. For further exploring the discrimination of each channel, we constructed a lightweight channel attention module by integrating ECA module after DW convolution in LW-SpaM. Then, we construct an MFFM to extract the fusion information of the spectral and spatial features extracted by SpeM and LW-SpaM for considering the strong complementarity related information among different hierarchical structures. Now we will illustrate the abovementioned parts in the following sections in detail.

B. Spectral Information Extraction Module (SpeM)

The 1D-CNN utilized in the early HSI classification task [16] only extracts limited spectral information, and the utilization of spatial information is scarce, usually resulting in poor classification results. Additionally, the computational complexity of 3-D PW convolution is relatively high; at the same time, 3-D PW convolution operation generates a lot of computational redundancy, which drags down the overall operating efficiency of the network. Therefore, different from aforementioned 1D-CNN and 3-D PW convolution, the proposed SpeM structure in this section is composed of multilayer PW convolutions, which can extract spectral information with less calculation. The function of SpeM is mainly focus on the fusion and reorganization of the weights of feature maps in the direction of the spectral dimension for constructing a distinguishable spectral feature vector, while avoiding interference of other pixel values around the sample to be classified (i.e., no spatial feature information is involved). Thereafter, the spatial information is extracted by LW-SpaM, that is, the SpeM and LW-SpaM are separated to generate better distinguishable spectral and spatial features for further fusion task.

As shown in Fig. 5, we extract the spectral features of the input feature map by multilayer PW convolutions, and we construct the PW convolution units (PW unit) in SpeM. PW unit is composed of PW convolution, ReLU6 nonlinear activation function, batch normalization (BN) layer, and residual connection. SpeM contains 18 PW units and 1 channel adjustment PW unit. The 18 PW units extract spectral information, and the channel adjustment PW unit adjusts the number of

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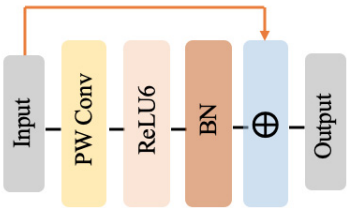


Fig. 5. Structure of PW unit.

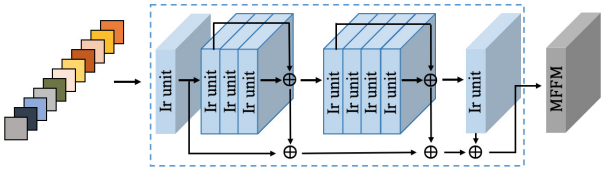


Fig. 6. LW-SpaM (in blue dotted line rectangle). Each Ir unit has the same number of channels, and the distribution ratio is 1:3:4:1.

output channels of SpeM and inputs it into LW-SpaM. Each PW unit is 64 channels. The channel adjustment PW unit is 32 channels.

C. Lightweight Spatial Information Extraction Module (LW-SpaM)

The LW-SpaM aims to extract the discriminative spatial features from the input feature map through the lightweight unit. The integral structure of LW-SpaM is constructed according to the NAS-guided conclusions and are presented in [47]. Most well-performing networks have the following three characteristics: 1) four-stage networks usually perform better than three-stage networks; 2) the models with good performance have a large number of units in the third stage and a small number of units in the last stage usually indicates better performance; 3) utilization of the inverted bottleneck (channel expansion rate $b < 1$) can degrade performance. According to the above conclusions, the structure of the LW-SpaM is shown in Fig. 6. The ratio of Ir units in each stage is 1:3:4:1. The channel expansion rate b is set to 1. The whole structure of LW-SpaM can be roughly divided into four stages. The basic unit of each stage is lightweight Ir units. The output of different stages is linearly mapped by PW convolution as the feature to be fused by MFFM so that the proposed model still retains relatively satisfying feature information in deeper network.

In the traditional MobileNet lightweight network, PW convolution consumes most of the overall calculation of the network, which mainly acts on the expansion and contraction of DW convolutional channels. To alleviate the above problems, we replaced the DW convolution in the original Ir unit with Ghost. Ghost reduces the percentage of PW convolution calculations in the overall network. In addition, Ghost can use multiple DW convolutions of different scales to extract the spatial features of the input feature map, so that more diverse and abstract feature information can be obtained. LW-SpaM mainly focuses on extracting spatial features. To make up the lack of attention for spectral information, we exploit the ECA module to adjust the channel weight and suppress the useless feature map. The ECA module is added after the DW convolution layer to adjust the channel weight. The input and output are connected with a residual connection if and only if they have the same number of channels. This structure maintains more distinguishable features at the input and output stages. If the number of input and output channels is the same, the structure is shown in Fig. 7.

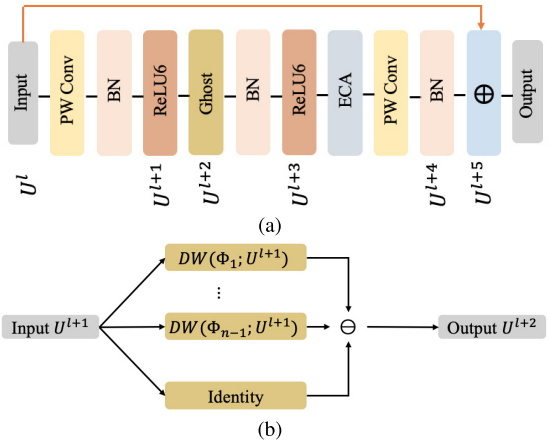


Fig. 7. Ir unit consists of two PW convolutions and one Ghost module, and the channel weight is adjusted through the Ghost module. (a) Ir unit overall structure. (b) Ghost internal structure. $\ominus$ stands for concat operation.

$$U^{l+1} = \varphi_{\mathrm{BN}}\left(\sigma\left(W^{l} \odot U^{l}\right)\right) \tag{12}$$

$$\hat{U}_i^{l+1} = \varphi_{\mathrm{DW}}\left(\Phi_i; U^{l+1}\right) i \in 1, 2, \cdots n-1 \tag{13}$$

$$U^{l+2} = \sigma\left(\varphi_{\mathrm{BN}}(\ominus(\hat{U}_1^{l+1}; \cdots; \hat{U}_{n-1}^{l+1}; U^{l+1}))\right) \tag{14}$$

$$U^{l+3} = \varphi_{\mathrm{ECA}}\left(W^{l+2}, U^{l+2}\right) \tag{15}$$

$$U^{l+4} = \varphi_{\mathrm{BN}}\left(\sigma\left(W^{l+3} \odot U^{l+3}\right)\right) \tag{16}$$

$$U^{l+5} = U^{l} \oplus U^{l+4} \tag{17}$$

where $\odot$ is the PW convolution operation, σ represents the ReLU6 nonlinear activation function, $\varphi_{\mathrm{DW}}(\Phi_i; .)$, φ_{BN}, φ_{ECA} indicates the DW convolution operation with Φ_i as the convolution kernel, BN and ECA module, respectively. The concat operation and residual connection are represented by $\ominus$ and $\oplus$, respectively.

The PW convolution is first utilized to reduce the channel number of input U^l to 32. Then, the spatial features in U^{l+1} are extracted through multiple DW convolutions and connected with the identity feature maps. Multiple DW convolutions can use different scale convolution kernels for input. Ghost uses two DW convolutions with 3×3 size. After incorporating ECA into Ghost, we adjust the channel weight to make up for the lack of attention to the spectrum in LW-SpaM. Finally, PW convolution is exploited to shrink the number of channels to 32. Shrinking is liable to cause information loss, thereby ReLU6 is no longer used here to prevent double information loss.

D. Multi-Layer Feature Fusion Module (MFFM)

The MFFM aims to consider the strong complementary information among different hierarchical structures and extracts the fusion information of the spectral and spatial features of SpeM and LW-SpaM. Thereby, in order to further reduce the network computation cost, different layer features

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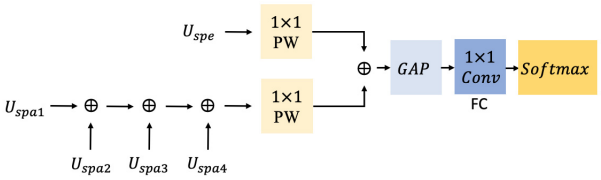


Fig. 8. MFFM fuses the spectral features and spatial features of different layers.

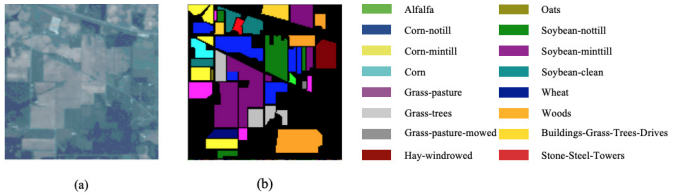


Fig. 9. IN. (a) False-color image. (b) Ground-truth map.

are dimensional-matched and interacted by PW convolution for the fusion of different feature maps in spectral and spatial feature extraction stage by addition operation.

To generate summary statistics of spectralwise $h^{\text{avg}} \in \mathbb{R}^{1\times1\times D}$, the GAP is operated on each fused spectral feature across spatial dimensions $H \times W$. Vector h^{avg} performs PW convolution and outputs $U^{\text{out}} \in \mathbb{R}^{1\times1\times C}$, where C is the number of pixel classes. The MFFM is shown in Fig. 8 and formulated as follows:

$$\hat{U}_i = W_i \odot U_{\text{spe1}} + b \tag{18}$$

$$U_{\text{spasum}} = F_{sum}\left(U_{\text{spa1}}; U_{\text{spa2}}; U_{\text{spa3}}; U_{\text{spa4}}\right) \tag{19}$$

$$\hat{U}_j = W_j \odot U_{\text{spasum}} + b \tag{20}$$

$$U_{\text{sum}} = \hat{U}_i \oplus \hat{U}_j \tag{21}$$

$$U^{\text{avg}} = \text{GAP}(U_{\text{sum}}) \tag{22}$$

$$U^{\text{out}} = W_{\text{avg}} \odot U^{\text{avg}} \tag{23}$$

where U_{spe1} represents spectral extraction feature, and U_{spai} represents multiple spatial extraction features. F_{sum} is the add operation. U_{spasum} denotes the spatial fusion feature. $\hat{U}_i$ and $\hat{U}_j$ are the adjusted features. GAP represents GAP operation. $\odot$ is the PW convolution operation. $U^{\text{out}} \in \mathbb{R}^{1\times1\times C}$ is the predicted label vector. The model is calculated by the function argmax as

$$\hat{Y} = \arg\max\left(U^{\text{out}}\right) \tag{24}$$

where $\arg\max(\cdot)$ represents the index of the maximum value in vector U^{out}. $\hat{Y}$ is the predicted class of the input 3-D patch cube data z_i.

E. Loss Function and Optimization

We adopt cross entropy as the loss function for classification of HSI in the proposed model. The cross entropy loss function is given as follows:

$$\text{Loss} = -\frac{1}{T}\sum_{t=1}^{T}\sum_{c=1}^{C} Y_c^t \log\left(\hat{Y}_c^t\right) \tag{25}$$

where C indicates the number of classes and Y_c^t and $\hat{Y}_c^t$ are the true and predicted labels, respectively. T is the number of samples in a minibatch. The parameters in the model are updated by backpropagation and stochastic gradient descent.

IV. Experiments and Discussion

We evaluate our proposed algorithm mainly on three real HSI data sets. Then the effect of parameters in the proposed model is analyzed. Finally, the comparison result with the existing algorithms are analyzed and discussed in this section.

A. Hyperspectral Data Sets

The Indian Pines (IN) data set was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the agricultural Indian Pine test site in northwestern Indiana in 1992. IN has 220 spectral dimension channels with wavelength ranging from 0.2 to 2.4 μm and 145 × 145 spatial dimension of 20 m/pixel. It has 16 different ground-truth classes with 10 249 labeled pixels. Fig. 9 shows the three-band false color composite of IN images and the corresponding ground-truth data, respectively. Before the experiment, 20 bands have been removed out due to the atmospheric water absorption.

The University of Pavia (UP) data set was acquired by the reflective optics system imaging spectrometer (ROSIS) sensor at the time of a flight campaign over the university campus at Pavia, Northern Italy in 2001. UP has 115 spectral dimension channels with wavelength ranging from 0.43 to 0.86 μm and 610 × 340 spatial dimension of 1.3 m/pixel. It has 9 different ground-truth classes with 42 776 labeled pixels. Fig. 10 shows the three-band false color composite of UP images and the corresponding ground-truth data, respectively. Before the experiment, 12 bands have been removed out due to noise.

The Houston University (HU) data set was gathered by compact airborne spectrographic imager (CASI) over the HU campus and the neighboring urban area in 2013. HU has 144 spectral dimension channels with wavelength ranging from 0.38 to 1.05 μm and 1905 × 349 spatial dimension of 2.5 m/pixel. It has 15 different ground-truth classes with 15 029 labeled pixels. Fig. 11 shows the three-band false color composite of HU images and the corresponding ground-truth data, respectively.

We choose different proportions of labeled samples as training samples, validation samples, and test samples to evaluate the performance of the proposed model. Therefore, we randomly select 3%, 1%, and 3% of the whole labeled samples as training sets, 10%, 5%, and 5% of the whole labeled samples as validation sets, and 97%, 99%, and 97% of the whole labeled samples as testing sets for the IN, UP, and HU data sets, respectively. The details of the number of each class for training, validation, and testing sets are listed in Tables II–IV.

B. Experimental Configuration and Preprocessing

The labeled 3-D patch cube data set is randomly divided into training, validation, and testing data sets represented by Z_{train}, Z_{val}, and Z_{test}, respectively, and corresponding label sets Y_{train}, Y_{val}, and Y_{test}, respectively. While using Z_{train} to optimize the model parameters, Z_{val} is used to verify the trained

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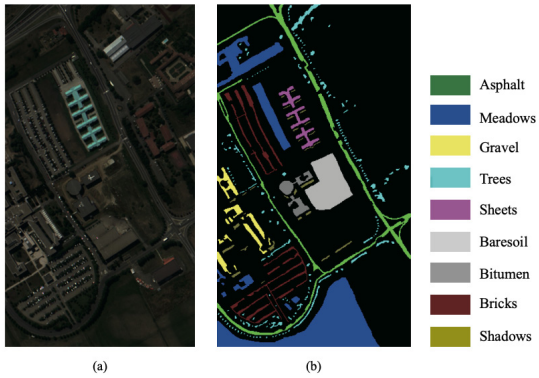


Fig. 10. UP. (a) False-color image. (b) Ground-truth map.

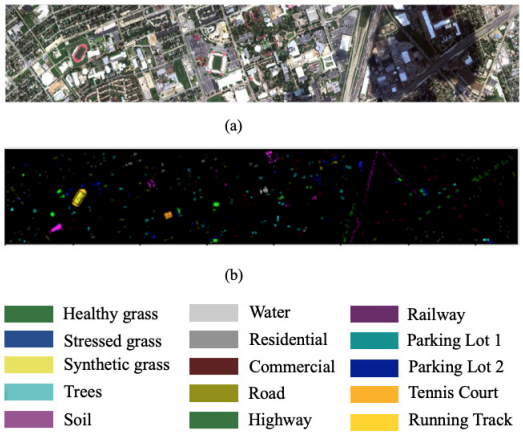


Fig. 11. HU. (a) False-color image. (b) Ground-truth map.

TABLE I

OA(%) FOR LMAFN USING ECA OR NOT. EX STANDS FOR USING ECA, DEX STANDS FOR NOT TO USE ECA

ECA	IN	UP	HU
EX	95.56 ± 0.85	98.59 ± 0.41	96.59 ± 0.88
DEX	94.98 ± 0.79	97.83 ± 0.37	96.16 ± 0.68

network. The parameters of a certain generation with the highest verification accuracy are saved, and Z_{test} is classified with the saved model parameters. Three evaluation metrics of performance are obtained by Z_{test} and all pixels are classified to construct a classification map.

In this article, Adam optimizer with the batch size of 32 is adopted. The learning rate controls the convergence speed in training progress and affects the convergence efficiency of the model. Here, we select the learning rate as 0.001 in our experiments. The computer configuration is as follows: An Intel Xeon W-2123 CPU at 3.60 GHz with 64-GB RAM and an NVIDIA GeForce GTX 2080 Ti graphical processing unit (GPU) with 11-GB RAM. The software environment is the system of 64-bit Windows 10 and deep learning frameworks of TensorFlow 1.14.0. Overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa) are utilized to evaluate the classification performance in this article.

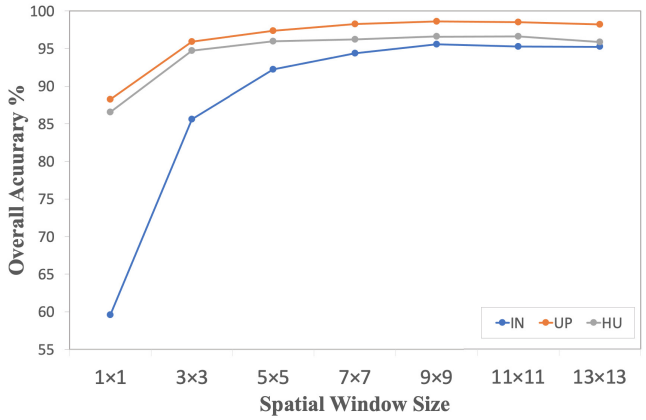


Fig. 12. OA(%) of LMAFN with different spatial input sizes in the IN, UP, and HU data sets.

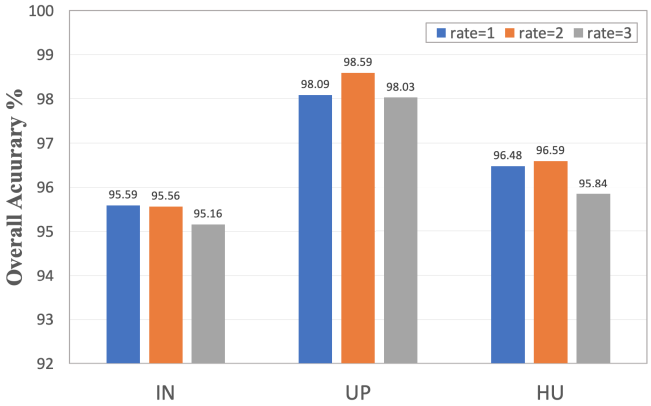


Fig. 13. OA(%) of LMAFN with the different channel multiplication rate of Ghost in the IN, UP, and HU data sets.

C. Parameter Setting and Analysis

In this part, we mainly analyze the parameters that affect the training process and classification performance of the proposed LMAFN, including spatial window size, the channel multiplication rate of Ghost, and the effectiveness of ECA. To avoid the inclination in random sampling, the evaluated mean $\pm$ std value is reported for parameter analysis and classification result comparison with ten times random repeated.

In the first set of experiments, the effect of the spatial window size is analyzed. Three-dimensional patch cubes contain spatial–spectral features. The spatial window size determines how much spatial information around the center pixel to be involved in the 3-D patch cube. To evaluate the influence of the spatial window size on the performance of LMAFN, we extract various spatial window sizes in 3-D patch cubes for three data sets. We select a varied set of spatial window sizes $\{1, 3, 5, 7, 9, 11, 13\}$ for IN, UP, and HU, respectively. In the process of evaluating the effect of spatial window size, we set the channel multiplication rate of Ghost as 2. The patch size analysis for three data sets is illustrated in Fig. 12. We observe that the most accurate patch choice is 9×9 for IN, UP, and HU.

The second set of experiments is executed to evaluate the channel multiplication rate of Ghost. The channel multiplication rate decides the feature representation capability, computational complexity, and module size. More channels

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TABLE II

CLASSIFICATION ACCURACY OF DIFFERENT METHODS FOR LABELED PIXELS OF THE IN DATA SET

Class	Class Name	Train	Val	Test	SVM	SVMCK	2D-CNN	3D-CNN	DFFN	SSRN	LMAFN
1	Alfalfa	2	5	44	51.15 ± 18.23	79.23 ± 6.82	78.79 ± 1.31	30.37 ± 2.57	$\mathbf{90.00 \pm 8.70}$	76.26 ± 20.00	83.18 ± 15.85
2	Corn-notill	43	143	1385	74.24 ± 4.41	83.60 ± 2.97	72.68 ± 1.45	89.58 ± 1.15	88.35 ± 3.54	$\mathbf{91.98 \pm 5.03}$	91.95 ± 3.40
3	Corn-mintill	25	83	805	61.61 ± 5.47	83.94 ± 5.53	77.31 ± 0.72	64.68 ± 1.71	87.17 ± 5.36	$\mathbf{93.07 \pm 4.21}$	90.52 ± 3.77
4	Corn	8	24	229	52.65 ± 8.77	79.65 ± 5.65	77.44 ± 2.41	53.19 ± 1.09	92.40 ± 4.30	77.47 ± 13.02	$\mathbf{93.01 \pm 5.07}$
5	Grass-pasture	15	49	468	87.55 ± 5.10	92.70 ± 2.95	81.41 ± 1.07	75.34 ± 0.12	87.99 ± 6.43	$\mathbf{99.75 \pm 0.35}$	90.60 ± 4.37
6	Grass-trees	22	73	708	93.67 ± 2.75	92.73 ± 4.14	85.12 ± 1.34	98.02 ± 0.24	91.45 ± 6.79	$\mathbf{98.39 \pm 1.47}$	98.32 ± 1.58
7	Grass-pasture-mowed	1	3	27	50.40 ± 32.56	95.20 ± 1.60	4.94 ± 2.14	44.44 ± 0.29	77.41 ± 23.68	86.32 ± 9.86	$\mathbf{90.00 \pm 16.81}$
8	Hay-windrowed	15	48	463	97.68 ± 1.23	97.59 ± 1.66	99.78 ± 0.08	100 ± 0	99.00 ± 1.54	97.41 ± 3.66	$\mathbf{99.09 \pm 0.96}$
9	Oats	1	2	19	58.95 ± 19.52	54.74 ± 21.98	68.42 ± 13.93	22.81 ± 6.08	81.58 ± 20.14	$\mathbf{93.94 \pm 8.57}$	67.37 ± 24.21
10	Soybean-nottill	30	98	942	63.18 ± 7.24	84.80 ± 4.55	79.26 ± 4.29	71.51 ± 0.58	88.41 ± 5.20	89.31 ± 11.89	$\mathbf{92.31 \pm 4.36}$
11	Soybean-minttill	74	246	2381	78.81 ± 2.29	89.39 ± 2.14	91.01 ± 1.24	90.54 ± 0.42	94.72 ± 1.06	91.27 ± 5.54	$\mathbf{95.30 \pm 1.55}$
12	Soybean-clean	18	60	575	67.16 ± 3.08	75.36 ± 8.85	88.99 ± 2.73	89.68 ± 1.23	90.52 ± 3.19	71.68 ± 11.96	$\mathbf{93.11 \pm 3.93}$
13	Wheat	7	21	198	98.05 ± 1.23	99.02 ± 0.62	96.30 ± 0.58	99.50 ± 0.47	90.51 ± 8.64	96.67 ± 2.89	$\mathbf{98.84 \pm 0.96}$
14	Woods	38	127	1227	94.12 ± 2.31	95.86 ± 1.62	89.95 ± 0.53	95.65 ± 0.05	96.81 ± 1.88	94.93 ± 3.12	$\mathbf{97.74 \pm 2.16}$
15	Buildings-Grass-Trees-Drives	12	39	374	41.58 ± 8.96	82.77 ± 6.47	88.32 ± 3.21	75.40 ± 0.80	90.27 ± 6.77	86.12 ± 8.92	$\mathbf{94.39 \pm 6.87}$
16	Stone-Steel-Towers	3	10	90	86.09 ± 7.64	95.22 ± 4.79	75.56 ± 4.84	72.22 ± 0.52	58.89 ± 8.37	93.33 ± 5.18	$\mathbf{95.22 \pm 6.07}$
	OA(%)				77.19 ± 1.00	88.17 ± 1.30	84.74 ± 0.30	85.40 ± 0.17	91.57 ± 1.35	89.98 ± 0.50	$\mathbf{94.37 \pm 0.71}$
	AA(%)				72.31 ± 2.12	86.36 ± 2.17	78.46 ± 1.31	73.31 ± 0.48	87.84 ± 3.33	89.87 ± 1.18	$\mathbf{91.94 \pm 3.15}$
	KAPPA(%)				73.92 ± 1.23	86.52 ± 1.48	82.55 ± 0.38	83.25 ± 0.20	90.38 ± 1.56	88.58 ± 0.51	$\mathbf{93.58 \pm 0.82}$

TABLE III

CLASSIFICATION ACCURACY OF DIFFERENT METHODS FOR LABELED PIXELS OF THE UP DATA SET

Class	Class Name	Train	Val	Test	SVM	SVMCK	2D-CNN	3D-CNN	DFFN	SSRN	LMAFN
1	Asphalt	67	332	6564	83.28 ± 2.02	92.82 ± 2.17	91.04 ± 2.77	96.81 ± 2.18	98.21 ± 1.29	99.04 ± 0.22	$\mathbf{99.14 \pm 0.72}$
2	Meadows	187	933	18462	95.53 ± 1.98	98.94 ± 0.49	97.49 ± 1.62	90.00 ± 15.18	99.47 ± 0.45	$\mathbf{99.54 \pm 0.24}$	99.42 ± 0.40
3	Gravel	21	105	2078	61.88 ± 5.55	86.40 ± 1.84	70.97 ± 8.52	86.15 ± 4.14	92.37 ± 5.47	$\mathbf{98.76 \pm 1.31}$	91.50 ± 4.23
4	Trees	31	154	3033	85.51 ± 3.77	93.60 ± 1.24	92.32 ± 3.46	93.40 ± 7.07	87.71 ± 2.56	$\mathbf{99.95 \pm 0.06}$	96.11 ± 1.35
5	Painted metal sheets	14	68	1331	98.55 ± 0.74	99.28 ± 0.34	99.22 ± 0.44	96.99 ± 4.96	95.18 ± 4.99	$\mathbf{99.95 \pm 0.04}$	99.80 ± 0.32
6	Bare Soil	51	252	4978	60.22 ± 4.85	93.98 ± 1.65	79.90 ± 7.80	89.64 ± 1.50	$\mathbf{99.40 \pm 0.93}$	98.79 ± 1.47	98.80 ± 1.13
7	Bitumen	14	67	1316	71.12 ± 6.36	90.62 ± 1.97	71.70 ± 6.69	88.18 ± 5.28	96.47 ± 3.16	$\mathbf{99.79 \pm 0.16}$	96.89 ± 2.58
8	Self_Blocking Bricks	37	185	3645	81.68 ± 2.90	93.08 ± 1.56	92.59 ± 4.63	87.03 ± 3.41	$\mathbf{96.83 \pm 2.54}$	88.54 ± 2.78	95.25 ± 1.96
9	Shadows	10	48	937	97.47 ± 2.15	87.92 ± 5.22	92.93 ± 2.35	99.75 ± 0.16	72.57 ± 5.23	97.22 ± 1.49	$\mathbf{99.18 \pm 0.81}$
	OA(%)				85.30 ± 0.86	95.41 ± 0.50	91.48 ± 2.10	94.21 ± 0.44	97.03 ± 0.70	98.26 ± 0.07	$\mathbf{98.27 \pm 0.49}$
	AA(%)				81.69 ± 1.56	92.96 ± 0.68	87.57 ± 1.97	91.99 ± 0.82	93.13 ± 1.21	$\mathbf{97.95 \pm 0.17}$	97.34 ± 0.75
	KAPPA(%)				80.21 ± 1.19	93.91 ± 0.66	88.62 ± 2.81	92.27 ± 0.59	96.06 ± 0.92	97.69 ± 0.09	$\mathbf{97.73 \pm 0.46}$

can extract more distinguishing features, but at the same time it will increase network running time and reduce efficiency. Therefore, we test the effects of different multiplication rates to achieve a balance between efficiency and accuracy. Here we set the spatial window size as 9×9 for IN, UP, and HU. In order to find an optimum multiplication rate, we verify the classification effect of the network under different multiplication rates $\{1, 2, 3\}$. The experimental result is presented in Fig. 13 that the best choice for the multiplication rate is 2 for IN, UP, and HU.

From above analysis of parameters in the proposed model, the spatial window size is set as 9×9 for IN, UP, and HU. Since IN, UP, and HU have highly distinguishable spectral features, we take all the spectral dimensions as input. It can be seen from Fig. 13 that the optimal channel multiplication rate of Ghost is 2 for IN, UP, and HU. Therefore, we set the channel multiplication rate to 2.

Theoretically, ECA can increase the feature discrimination between channels by adjusting the weight value of each channel. ECA sets up a mapping relationship to adapt the cross-channel interaction range according to the number of channels. In the process of evaluating the effectiveness of ECA, we set the channel multiplication rate of Ghost as 2, and the spatial window size as 7×7 for IN, UP, and HU. As shown in Table I, by using the ECA module to adjust channel-wise features, the network learns to use global feature information to selectively enhance effective features and suppress useless features. We can observe that ECA can improve 0.5%–1% accuracy in three data sets, which improves the classification accuracies without introducing extra computation burden for three data sets.

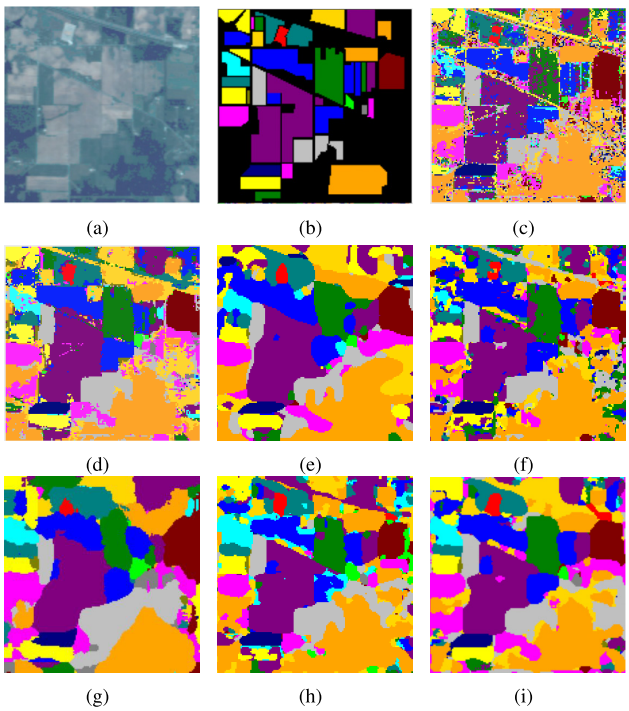


Fig. 14. Classification maps for IN. (a) False-color image. (b) Ground-truth map. (c) SVM. (d) SVMCK. (e) 2D-CNN. (f) 3D-CNN. (g) DFFN. (h) SSRN. (i) LMAFN.

D. Classification Results and Analysis

In this section, we compare the proposed LMAFN model with two traditional methods: SVM and extended SVMCK, and four deep learning-based methods: 2D-CNN, 3D-CNN, DFFN, and SSRN. The quantitative comparison results of

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TABLE IV
CLASSIFICATION ACCURACY OF DIFFERENT METHODS FOR LABELED PIXELS OF THE HU DATA SET

Class	Class Name	Train	Val	Test	SVM	SVMCK	2D-CNN	3D-CNN	DFFN	SSRN	LMAFN
1	Healthy grass	38	63	1213	96.78 ± 2.11	97.16 ± 1.74	90.74 ± 2.19	96.37 ± 0.59	93.62 ± 2.32	$\mathbf{97.97 \pm 2.37}$	97.54 ± 2.47
2	Stressed grass	38	63	1216	96.38 ± 1.69	95.99 ± 3.32	87.21 ± 3.01	96.38 ± 1.08	92.80 ± 3.11	95.34 ± 4.60	$\mathbf{98.60 \pm 1.30}$
3	Synthetic grass	21	35	676	99.70 ± 0.09	99.62 ± 0.48	96.38 ± 0.60	98.67 ± 1.31	97.91 ± 2.32	$\mathbf{100 \pm 0}$	99.78 ± 0.39
4	Trees	38	63	1206	95.14 ± 2.48	92.02 ± 4.06	93.93 ± 0.84	96.85 ± 3.74	85.61 ± 3.77	$\mathbf{99.66 \pm 0.32}$	98.26 ± 1.83
5	Soil	38	63	1204	98.70 ± 0.56	98.32 ± 1.43	97.88 ± 0.69	99.72 ± 0.19	99.57 ± 0.98	95.46 ± 2.30	$\mathbf{99.65 \pm 0.62}$
6	Water	10	17	315	89.65 ± 5.23	90.35 ± 4.07	70.39 ± 1.86	97.04 ± 2.88	88.70 ± 5.08	$\mathbf{100 \pm 0}$	93.08 ± 5.00
7	Residential	39	64	1229	84.38 ± 2.59	91.29 ± 4.07	95.09 ± 1.82	86.37 ± 3.48	92.70 ± 5.25	94.68 ± 2.33	$\mathbf{94.85 \pm 2.77}$
8	Commercial	38	63	1206	78.18 ± 7.29	86.98 ± 2.43	76.07 ± 4.30	73.76 ± 3.27	87.31 ± 5.20	$\mathbf{97.48 \pm 1.42}$	90.37 ± 2.43
9	Road	38	63	1214	75.11 ± 9.71	88.90 ± 6.12	86.43 ± 2.82	79.63 ± 0.96	87.95 ± 3.65	$\mathbf{94.76 \pm 0.91}$	89.77 ± 2.82
10	Highway	37	62	1190	87.13 ± 2.57	89.53 ± 2.77	85.53 ± 4.76	84.52 ± 7.45	97.38 ± 1.67	90.60 ± 3.71	$\mathbf{97.12 \pm 1.60}$
11	Railway	38	62	1197	78.58 ± 5.33	86.70 ± 2.01	82.85 ± 1.87	88.62 ± 4.95	96.54 ± 3.13	85.88 ± 6.22	$\mathbf{96.56 \pm 2.31}$
12	Parking Lot 1	37	62	1196	80.47 ± 5.80	88.95 ± 2.18	86.18 ± 3.59	86.32 ± 2.28	92.58 ± 2.63	90.77 ± 7.53	$\mathbf{97.43 \pm 1.62}$
13	Parking Lot 2	15	24	454	30.00 ± 4.72	76.56 ± 2.37	89.96 ± 0.67	76.41 ± 13.92	93.37 ± 5.23	$\mathbf{94.50 \pm 0.78}$	92.56 ± 4.09
14	Tennis Court	13	22	415	96.24 ± 5.28	97.93 ± 1.89	92.31 ± 0.57	95.74 ± 5.75	99.93 ± 2.40	$\mathbf{100 \pm 0}$	98.10 ± 2.45
15	Running Track	20	33	640	98.53 ± 0.53	99.88 ± 0.17	83.94 ± 3.06	98.91 ± 1.09	97.39 ± 2.93	98.33 ± 1.37	$\mathbf{99.92 \pm 0.23}$
	OA(%)				86.72 ± 2.07	92.02 ± 0.81	88.19 ± 0.27	89.76 ± 0.65	93.20 ± 0.58	94.59 ± 0.32	$\mathbf{96.24 \pm 0.66}$
	AA(%)				85.66 ± 1.97	92.01 ± 0.82	87.66 ± 0.37	90.37 ± 1.09	93.56 ± 0.60	95.70 ± 0.17	$\mathbf{96.24 \pm 0.73}$
	KAPPA(%)				85.63 ± 2.24	91.37 ± 0.87	87.22 ± 0.30	88.93 ± 0.71	92.65 ± 0.62	94.16 ± 0.35	$\mathbf{95.94 \pm 0.72}$

TABLE V
PARAMETER, DEPTH, OA AND DETAILED RUNNING TIME OF MODELS BASED ON DEEP LEARNING METHODS FOR THREE DATA SETS

Dataset	IN			UP			HU		
Model	Patameter	Depth	OA(%)	Patameter	Depth	OA(%)	Patameter	Depth	OA(%)
2D-CNN	186096	3	84.74	185193	3	91.48	185967	3	88.19
3D-CNN	9068	3	85.40	5253	3	94.21	8523	3	89.76
DFFN	374880	27	91.57	443929	33	97.03	375103	27	91.96
SSRN	376892	13	89.98	229261	13	98.26	290851	13	94.59
LMAFN	148651	57	94.37	140260	57	98.27	143658	57	96.24

different methods are shown in Tables II–IV. Compared with the traditional method SVM, SVMCK and deep learning-based algorithms can achieve better classification results on three data sets. This is because SVM only uses spectral features for classification, ignoring important spatial features. SVMCK combines spectral features and spatial features based on SVM, and its classification performance has been greatly improved compared with SVM. Through convolution operation and nonlinear activation functions, the CNN-based methods can learn more discriminative spectral–spatial features and map the features into more abstract high-level features. The CNN-based methods are divided into 2-D-based CNNs and 3-D-based CNNs. Compared with 2-D-based CNNs, 3-D CNNs can fully learn spectral features, thereby achieving better classification accuracy results and performance. SSRN has the highest classification accuracy among most comparison algorithms. On the three data sets, we can observe from Tables II–IV, the proposed LMAFN based on 2D-CNN is better than SSRN based on 3D-CNN. This is because the proposed LMAFN adopts lightweight structures, which greatly increases the network depth while reducing the size of the model as well as enhances the nonlinear fitting ability of the model. This allows the model to learn more abstract and differentiated advanced features. At the same time, the model uses the ECA attention module to enhance useful spectral bands and suppress useless spectral with extracting better spectral features. Therefore, the proposed LMAFN achieves the best accuracy among all comparison algorithms.

Benefit from the utilization of lightweight structures, LMAFN maintains a small model size while greatly increasing the depth of the model. Table V compares the parameter, network depth, and classification performance of 2D-CNN, 3D-CNN, DFFN, SSRN, and LMAFN on the three data sets. The proportion of training samples is the same as Tables II–IV. From Table V, based on the UP data set, we can notice that LMAFN has only 140 260 parameters, which is 68.4% less than 443 929 parameters of DFFN and 38.8% less than 229 261 parameters of SSRN. While reducing the size of the model, LMAFN increases the network depth to 57 layers and presents the most satisfying accuracies for three different data sets. The increase in depth enhances the nonlinear fitting ability of the model, so that the model can extract high-dimensional abstract spectral–spatial features.

TABLE VI
RUNNING TIME FOR THREE DATA SETS, TRAINING REFERS TO TOTAL TRAINING TIME. TEST REPRESENTS THE TIME TO TOTAL TEST TIME (IN SECONDS)

	IN		UP		HU	
	Training	Test	Training	Test	Training	Test
DFFN	337.72	1.10	376.53	3.98	350.40	1.54
SSRN	277.34	10.47	290.77	27.38	350.66	11.55
LMAFN	171.23	0.82	156.43	1.73	161.13	0.83

We also list the running time of LMAFN for each data set in Table VI. The total training time and the test time are reported, respectively. We can observe that proposed LMAFN only cost 171.23, 156.43, and 161.43 s for training, and 0.82, 1.73, and 0.83 s for testing for three data sets, while in practical remote

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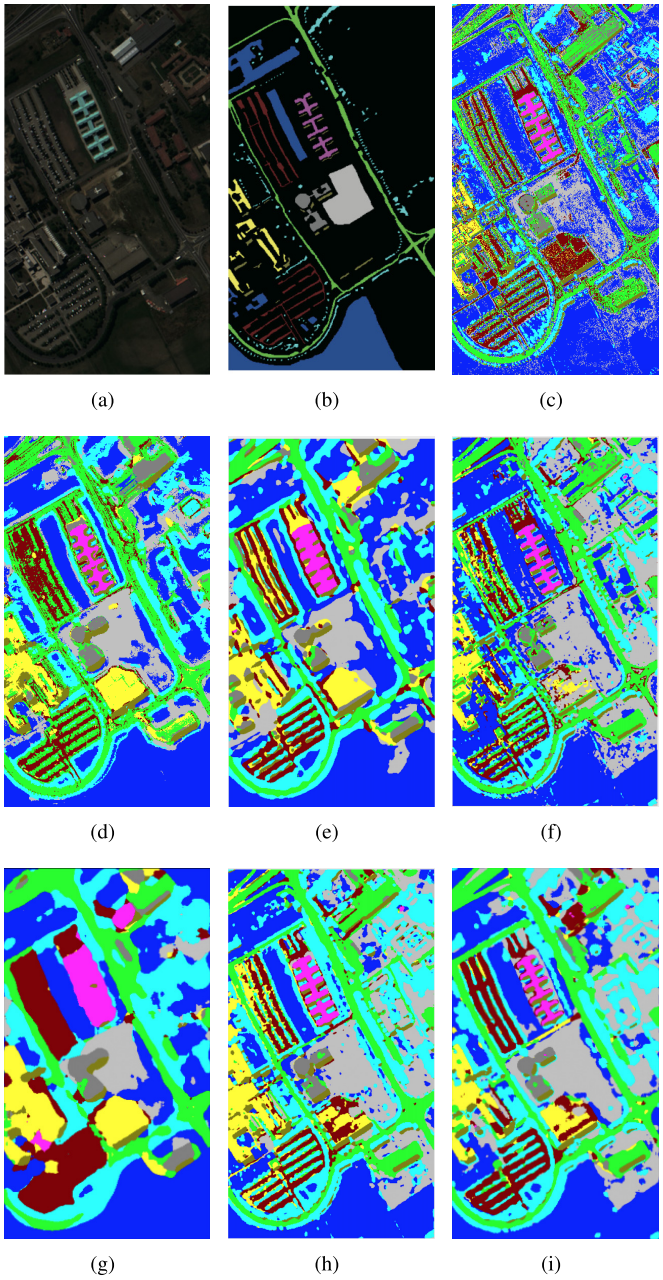


Fig. 15. Classification maps for UP. (a) False-color image. (b) Ground-truth map. (c) SVM. (d) SVMCK. (e) 2D-CNN. (f) 3D-CNN. (g) DFFN. (h) SSRN. (i) LMAFN.

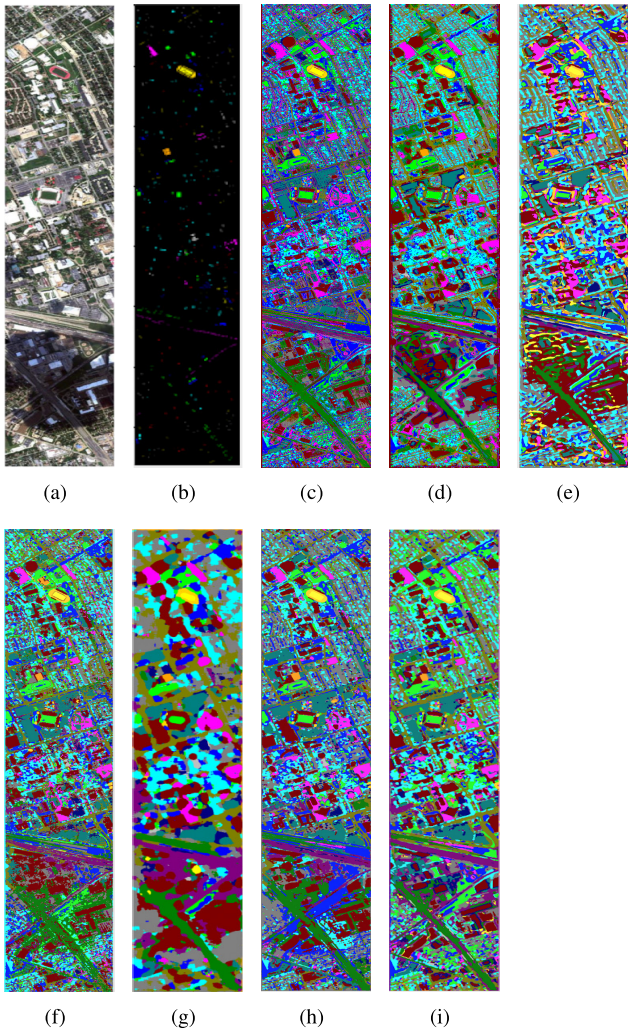


Fig. 16. Classification maps for HU. (a) False-color image. (b) Ground-truth map. (c) SVM. (d) SVMCK. (e) 2D-CNN. (f) 3D-CNN. (g) DFFN. (h) SSRN. (i) LMAFN.

sensing applications, the training process is done offline. The test time for the different data sets, such as the HU data set, it consists of 349×1905 pixels with 2.5 m spatial resolution, our method can process such a large data sets just in 0.83 s for test. Therefore, from Tables V and VI, the experimental results further demonstrate that LMAFN can increase the depth of the network while reducing the parameters with more satisfying classification accuracies and execution time.

The comparison results of classification maps for different algorithms are presented in Figs. 14–16. We can observe that the classification map of SVM that only exploits spectral features for classification, which presents the most random misclassified noise labels on three data sets. Spectral–spatial features are exploited in SVMCK, 2D-CNN, 3D-CNN, DFFN, and SSRN, and therefore the classification maps present more smooth results than SVM. In addition, compared with other comparison algorithms, SSRN presents smooth results and high accuracies because of simultaneously considering spatial and continuous spectral features. Compared with the above-mentioned comparison algorithms, the proposed LMAFN still illustrates the most satisfying accuracies and results with deeper structure and less model parameters.

Further evaluating the classification effect on different sample sizes, we conducted the experiments with different proportions of training samples based on three data sets, that is, 1%, 2%, 3%, 4%, and 5% of labeled pixels as the training set to train the LMAFN. The spatial window size and the channel multiplication rate are 9×9 and 2, respectively. The classification results over IN, UP, and HU data sets are summarized in Fig. 17. As shown in Fig. 17, we can observe that the OA increases as the increased proportion of training samples for all three HSI data sets. Our proposed algorithm presents more satisfying results with deeper network structure as well as less parameters cost.

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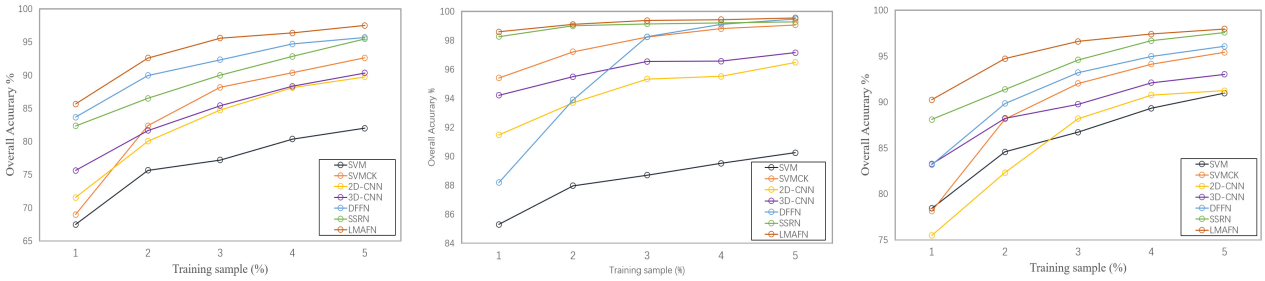


Fig. 17. Overall Accuracy (OA) % with different training rate with 1%, 2%, 3%, 4%, and 5% for (a) Indian Pines (IN), (b) Pavia University (UP), and (C) Houston University (HU).

V. Conclusion

In this article, a lightweight LMAFN deep learning network framework for HSI classification is proposed. LMAFN learns spectral and spatial features through lightweight structure under the constructive conclusions of NAS, and the SpeM and the LW-SpaM are constructed according to the conclusions of NAS presented in [47]. Therefore, in the process of spatial feature extraction, multiscale features are extracted through Ghost, and spectral features are recalibrated the weight through ECA attention module to make the weight of invalid feature map close to zero and increase the weight of the effective feature map. At the fusion stage, the strong complementary information among different hierarchical structures are extracted and interacted by PW convolution to make full use of the feature information between different layers for further improving the classification effect. Generally, the proposed LMAFN framework is an efficient lightweight deep network architecture and can achieve satisfying classification accuracies and performance with less parameters, computation resources even under fewer labeled training samples. The classification results of different data sets, different spatial window sizes, and different proportions of training samples confirm the feature generalization ability of the proposed LMAFN.

The future direction of our work is to further maintain the network classification performance by designing more efficient lightweight network framework through automatic architecture search algorithm.

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Jianing Wang (Member, IEEE) received the B.S. and M.S. degrees in circuit and system from Lanzhou University, Lanzhou, China, in 2005 and 2008, respectively, and the Ph.D. degree from Xidian University, Xi'an, China, in 2016.

She worked with China Aerospace Science and Technology Corporation, Xian. She is an Associate Professor and a member of the School of Artificial Intelligence, Xidian University, and the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China, Xidian University. Her research interests include image processing, machine learning, multiscale geometric analysis, compressive sampling, and artificial intelligent algorithm and applications.



Runhu Huang received the bachelor's degree from the School of Intelligence Science and Technology, Xidian University, Xi'an, China, in 2020.

He is a Graduate Student with the School of Artificial Intelligence, Xidian University. His research interests include model compression, deep learning, and remote sensing image processing.



Siying Guo received the bachelor's degree from the School of Computer Science and Technology, Xidian University, Xi'an, China, in 2020.

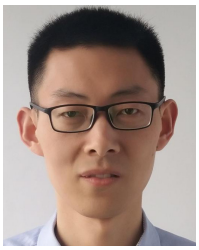
She is a Graduate Student with the School of Artificial Intelligence, Xidian University. Her research interests include deep learning, remote sensing data processing, and image processing.



Linhao Li received the bachelor's degree from the School of Physics and Optoelectronic Engineering, Xidian University, Xi'an, China, in 2019.

He is a Graduate Student with the School of Artificial Intelligence, Xidian University. His research interests include hyperspectral image processing, machine learning, and deep learning.

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Minghao Zhu received the B.S. degree from Anyang Normal University, Anyang, China, in 2008, and the M.S. degree from the Xi'an University of Science and Technology, Xi'an, China, in 2011. He is pursuing the Ph.D. degree in pattern recognition and intelligent systems with Xidian University, Xi'an.

His research interests include visual attention, machine learning, deep learning, and remote sensing image processing.



Shuyuan Yang (Senior Member, IEEE) received the B.S. degree in electrical engineering and the M.S. and Ph.D. degrees in circuits and systems from Xidian University, Xi'an, China, in 2000, 2003, and 2005, respectively.

She is a Full Professor with Xidian University. Her research interests include machine learning, computer vision, and compressive sampling.



Licheng Jiao (Fellow, IEEE) received the B.S. degree in high voltage from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees in electronic engineering from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

From 1984 to 1986, he was an Assistant Professor with the Civil Aviation Institute of China, Tianjin, China. From 1990 to 1991, he was a Post-Doctoral Fellow with the Key Laboratory for Radar Signal Processing, Xidian University, Xi'an, where he is the Director of the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education of China. He has authored or coauthored more than 200 scientific articles. His research interests include signal and image processing, nonlinear circuits and systems theory, wavelet theory, natural computation, and intelligent information processing.

Dr. Jiao is also a member of the IEEE Xian Section Executive Committee and an Executive Committee Member of the Chinese Association of Artificial Intelligence. He is also the Chairman of the Awards and Recognition Committee.

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